

27. Assuming the first 2 rows have been selected, how many ways are there to choose the 2 people in the back row?
28. Using the Multiplication Rule for Counting and your answers to the earlier parts of this exercise, how many ways are there for your friends to sort yourselves into rows to board the roller coaster?

The University Combinatorics Club has 18 members. Four of them will be selected to form a committee.

29. How many different committees of 4 are possible, assuming all of the duties are shared equally?
30. Instead of sharing responsibility equally, one person will be chosen to be the committee chair. How many different committees are possible? Count these by selecting a chair first, then selecting the remaining 3 members of the committee from the remaining club members and use the Multiplication Rule for Counting. Show your work.
31. Let's count the number of committees with chairs a different way: First, choose 4 people for the committee (as in the first question), then choose 1 of the 4 to be chair. Show your work. Do you get the same number?

Powerball® is a multistate lottery game, which costs \$2 to play. Players fill out a ticket by choosing 5 numbers between 1 and 69 (these are the white balls) and then a single number between 1 and 26 (this is the Powerball).

32. How many different ways are there to choose the white balls? Players who match these 5 numbers exactly (but not the Powerball) win \$1 million.
33. How many ways are there to choose the Powerball? Players who correctly pick the Powerball win \$4.
34. How many ways are there to play the game altogether? Players who match all 5 white balls and the Powerball win (or share) the grand prize. (The grand prize starts at \$40 million; if no players win the grand prize, the value goes up for the next drawing. The highest value it has ever reached is \$1.586 *billion*!) How many ways are there to fill out a single Powerball ticket?
35. You are in charge of programming for a music festival. The festival has a main stage, a secondary stage, and several smaller stages. There are 40 bands confirmed for the festival. Five of those will play the main stage, and 8 will play the secondary stage. How many ways are there for you to allocate bands to these 2 stages?

7.4 Tree Diagrams, Tables, and Outcomes



Figure 7.10 In genetics, the characteristics of an offspring organism depends on the characteristics of its parents. (credit: "Pea Plant" by Maria Keays/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Determine the sample space of single stage experiment.
2. Use tables to list possible outcomes of a multistage experiment.
3. Use tree diagrams to list possible outcomes of a multistage experiment.

In the 19th century, an Augustinian friar and scientist named Gregor Mendel used his observations of pea plants to set out his theory of genetic propagation. In his work, he looked at the offspring that resulted from breeding plants with different characteristics together. For applications like this, it is often insufficient to only know in how many ways a

process might end; we need to be able to list all of the possibilities. As we've seen, the number of possible outcomes can be very large! Thus, it's important to have a strategy that allows us to systematically list these possibilities to make sure we don't leave any out. In this section, we'll look at two of these strategies.

Single Stage Experiments

When we are talking about combinatorics or probability, the word “**experiment**” has a slightly different meaning than it does in the sciences. Experiments can range from very simple (“flip a coin”) to very complex (“count the number of uranium atoms that undergo nuclear fission in a sample of a given size over the course of an hour”). Experiments have unknown outcomes that generally rely on something random, so that if the experiment is repeated (or **replicated**) the outcome might be different. No matter what the experiment, though, analysis of the experiment typically begins with identifying its sample space.

The **sample space** of an experiment is the set of all of the possible outcomes of the experiment, so it's often expressed as a set (i.e., as a list bound by braces; if the experiment is “randomly select a number between 1 and 4,” the sample space would be written $\{1, 2, 3, 4\}$).

EXAMPLE 7.12

Finding the Sample Space of an Experiment

For each of the following experiments, identify the sample space.

1. Flip a coin (which has 2 faces, typically called “heads” and “tails”) and note which face is up.
2. Flip a coin 10 times and count the number of heads.
3. Roll a 6-sided die and note the number that is on top.
4. Roll two 6-sided dice and note the sum of the numbers on top.

Solution

1. If we use “H” to denote “heads is facing up” and “T” to denote “tails is facing up”, then the sample space is $\{H, T\}$.
2. It's possible (though unlikely) that there will be no heads flipped; the outcome in that case would be “0.” It's also possible (more likely, but still quite unlikely) that only one flip will result in heads. Any other whole number is possible, up to the maximum: We're flipping the coin 10 times, so we can't get any more than 10 heads. So, the sample space is $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.
3. There are 6 numbers on the die: 1, 2, 3, 4, 5, and 6. So, the sample space for a single roll of the die is $\{1, 2, 3, 4, 5, 6\}$.
4. If we roll 2 dice, the smallest possible sum we could get is $1 + 1 = 2$ and the biggest is $6 + 6 = 12$. Every other whole number between those two is possible. So, the sample space is $\{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$.

YOUR TURN 7.12

Identify the sample space of each experiment.

1. You draw a card from a standard deck and note its suit.
2. You draw a card from a standard deck and note its rank.
3. You roll a 4-sided die and note the number on the bottom. (A 4-sided die is shaped like a pyramid, so when it comes to rest, there's no single side facing up.)
4. You roll three 4-sided dice and note the sum of the numbers on the bottom.

Multistage Experiments

Some experiments have more complicated sample spaces because they occur in stages. These stages can occur in succession (like drawing cards one at a time) or simultaneously (rolling 2 dice). Sample spaces get more complicated as the complexity of the experiment increases, so it's important to choose a systematic method for identifying all of the possible outcomes. The first method we'll discuss is the table.

Using Tables to Find Sample Spaces

Tables are useful for finding the sample space for experiments that meet two criteria: (1) The experiment must have only two stages, and (2) the outcomes of each stage must have no effect on the outcomes of the other. When the stages do not affect each other, we say the stages are **independent**. Otherwise, the stages are **dependent** and so we can't use tables; we'll look at a method for analyzing dependent stages soon.

EXAMPLE 7.13**Determining Independence**

Decide whether the two stages in these experiments are independent or dependent.

1. You flip a coin and note the result, and then flip the coin again and note the result.
2. You draw 2 cards from a standard deck (52 cards), one at a time.

✓ **Solution**

1. No matter what happens on the first flip, the second flip has the same sample space: {H, T} (You'll sometimes hear the phrase "The coin has no memory"). So, these stages are independent.
2. Let's say that the first card you draw is A♠. The sample space for the second draw consists of all the cards *except* A♠ (since that card is no longer in the deck, you can't draw it again). If instead that first card was 2♥, the sample space for the second draw is different: it's every card except 2♥. Since the sample space for the second card changes based on the result of the first draw, these stages are dependent.

> **YOUR TURN 7.13**

Decide whether the two stages in these experiments are dependent or independent.

1. You're getting dressed to go to a party, and you plan to wear a blouse and a skirt. You choose the blouse first, then the skirt (assume that you'd be comfortable wearing any of your skirts with any of your blouses).
2. On further reflection, you realize that some of your skirts clash with some of your blouses. So, you choose the blouse first, and then choose a skirt that goes with your chosen blouse.

If you have a two-stage experiment with independent stages, a table is the most straightforward way to identify the sample space. To build a table, you list the outcomes of one stage of the experiment along the top of the table and the outcomes of the other stage down the side. The cells in the interior of the table are then filled using the outcomes associated with each cell's row and column. Let's look at an example.

EXAMPLE 7.14**Using Tables to Identify Sample Spaces**

Identify the sample spaces of these experiments using tables.

1. You roll two dice: one 4-sided and one 6-sided.
2. You're in an ice-cream shop, and you're going to get a single scoop of ice cream with a topping. The flavors of ice cream you're considering are vanilla, chocolate, and rocky road; the toppings are fudge, whipped cream, and sprinkles.
3. The pea plants you're breeding have two possible pod colors: green and yellow. These colors are decided by a particular gene, which comes in two types: "G" for green, and "g" for yellow (In genetics, capital letters usually denote *dominant* genes, while lower-case letters denote *recessive* genes). Each plant has two genes. If you breed a Gg pea plant with a gg plant, the offspring plant will get one gene from each parent. What are the possible outcomes?

✓ **Solution**

1. **Step 1:** Make the outline of a table, with the results of the 4-sided roll on one side and the results of the 6-sided roll on the other. In practice, it doesn't matter which you choose; for this example, we'll put the 4-sided results on top (labeling the columns of the chart) and the 6-sided results on the side (labeling the rows of the chart) as shown in the following table:

		4-Sided Roll			
		1	2	3	4
6-Sided Roll	1				
	2				
	3				
	4				
	5				
	6				

Step 2: Fill in the results in each cell of the table below. Use the notation of an ordered pair, with the row label first.

		4-Sided Roll			
		1	2	3	4
6-Sided Roll	1	(1,1)	(1,2)	(1,3)	(1,4)
	2	(2,1)	(2,2)	(2,3)	(2,4)
	3	(3,1)	(3,2)	(3,3)	(3,4)
	4	(4,1)	(4,2)	(4,3)	(4,4)
	5	(5,1)	(5,2)	(5,3)	(5,4)
	6	(6,1)	(6,2)	(6,3)	(6,4)

So, for example, (3,2) represents the outcome where the 6-sided roll results in a 3, and the 4-sided roll gives us a 2. Thus, the sample space of the experiment is $\{(1,1), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), (2,4), (3,1), (3,2), (3,3), (3,4), (4,1), (4,2), (4,3), (4,4), (5,1), (5,2), (5,3), (5,4), (6,1), (6,2), (6,3), (6,4)\}$.

2. **Step 1:** Let's put the flavors on the rows and the toppings on the columns of the following table:

		Toppings		
		fudge	whipped cream	sprinkles
Flavors	vanilla			
	chocolate			
	rocky road			

Step 2: We can fill in the cells of the table below with the resulting combinations.

Toppings				
		fudge	whipped cream	sprinkles
Flavors	vanilla	vanilla with fudge	vanilla with whipped cream	vanilla with sprinkles
	chocolate	chocolate with fudge	chocolate with whipped cream	chocolate with sprinkles
	rocky road	rocky road with fudge	rocky road with whipped cream	rocky road with sprinkles

So, the sample space is {vanilla with fudge, vanilla with whipped cream, vanilla with sprinkles, chocolate with fudge, chocolate with whipped cream, chocolate with sprinkles, rocky road with fudge, rocky road with whipped cream, rocky road with sprinkles}.

3. **Step 1:** We'll put the parents' genes (P1 and P2) as labels on the rows and columns of the following table:

		P2	
		g	g
P1	G		
	g		

Step 2: We'll fill in the offspring's gene composition, listing parent 1's gene first in the table below.

		P2	
		g	g
P1	G	Gg	Gg
	g	gg	gg

Thus, the sample space is {Gg, Gg, gg, gg}. (Diagrams like this, which allow us to identify the genotypes of offspring, are called *Punnett squares* in honor of Reginald Punnett (1875–1967), who first used them in the context of genetics.)

> YOUR TURN 7.14

1. Use a table to identify the sample space of an experiment in which you flip a coin and roll a 6-sided die.

Using Tree Diagrams to Identify Sample Spaces

In experiments where there are more than two stages, or where the stages are dependent, a **tree diagram** is a helpful tool for systematically identifying the sample space. Tree diagrams are built by first drawing a single point (or *node*), then from that node we draw one branch (a short line segment) for each outcome of the first stage. Each branch gets its own node at the other end (which we typically label with the corresponding outcome for that branch); from each of these, we draw another branch for each outcome of the second stage, assuming that the outcome of the first stage matches the branch we were on. If there are other stages, we can continue from there by continuing to add branches and nodes. This sounds really complicated, but it's easier to understand through an example.

EXAMPLE 7.15**Using a Tree Diagram to Identify a Sample Space**

Use a tree diagram to find the sample spaces of each of the following experiments:

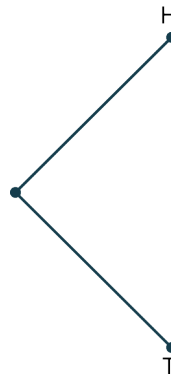
1. You flip a coin 3 times, noting the outcome of each flip in order.
2. You flip a coin. If the result is heads, you roll a 4-sided die. If it's tails, you roll a 6-sided die.
3. You are planning to go on a hike with a group of friends. There are 3 trails to consider: Abel Trail, Borel Trail, and Condorcet Trail. One of your friends, Jess, requires a wheelchair; if she joins you, the group couldn't handle the rocky Condorcet Trail.

✓ Solution

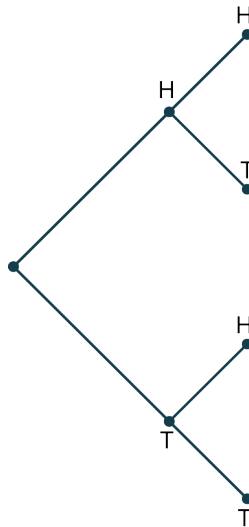
1. **Step 1:** Let's start by placing our first node ([Figure 7.11](#)).

**Figure 7.11**

- Step 2:** We'll add two branches, one for each outcome of the first coin flip, and label them ([Figure 7.12](#)).

**Figure 7.12**

- Step 3:** We're ready for stage two of the experiment: another coin flip. At each node, we add in branches that represent those outcomes ([Figure 7.13](#)).

**Figure 7.13**

Finally, we can add another set of branches for the outcomes of the third stage ([Figure 7.14](#)).

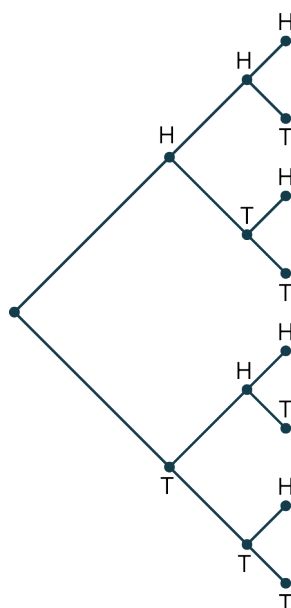


Figure 7.14

(These final nodes are called *leaves*.)

Step 4: We can write down the outcomes in the sample space by tracing the path out to each leaf, writing down the outcome at each node we pass through. For example, this leaf ([Figure 7.15](#)):

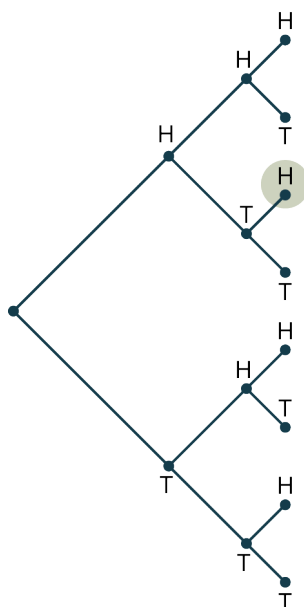


Figure 7.15

is reached via this path ([Figure 7.16](#)):

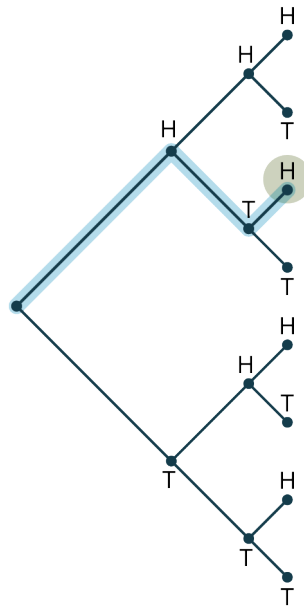


Figure 7.16

Step 5: We'll label that leaf as "HTH" (Figure 7.17), since the path passes through nodes labeled H, T, and H on its way out to our leaf.

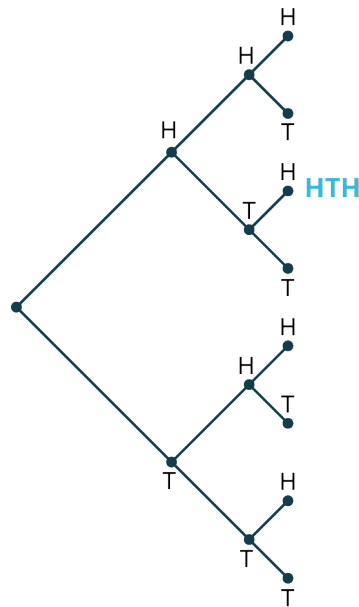


Figure 7.17

Step 6: We can label the remaining leaves using the same method (Figure 7.18).

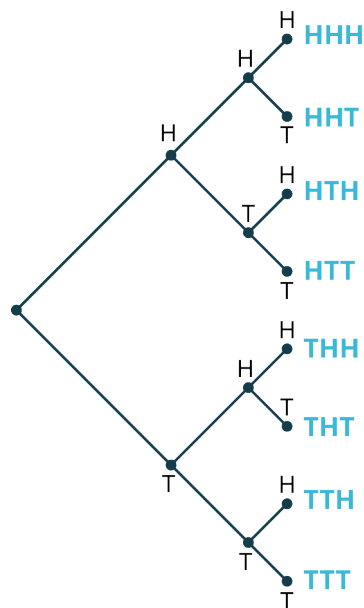


Figure 7.18

The sample space is the labels on the leaves: {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}.

2. **Step 1:** We'll start with our initial node ([Figure 7.19](#)).



Figure 7.19

Step 2: We'll add in branches for the outcomes of the first stage ([Figure 7.20](#)), which is the coin flip.

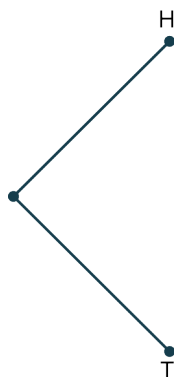


Figure 7.20

Step 3: The second stage of the experiment depends on the outcome of the first stage.

- a. If the outcome of the first stage was H, then we roll a 4-sided die. So, *only* on the node for H, we'll add in the outcomes of a 4-sided die roll ([Figure 7.21](#)).

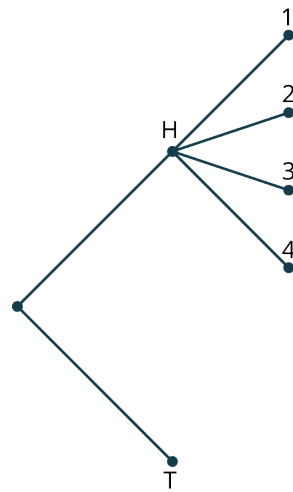


Figure 7.21

- b. If the outcome of the first stage was T, then we roll a 6-sided die. So, we'll add those branches to the node for T ([Figure 7.22](#)).

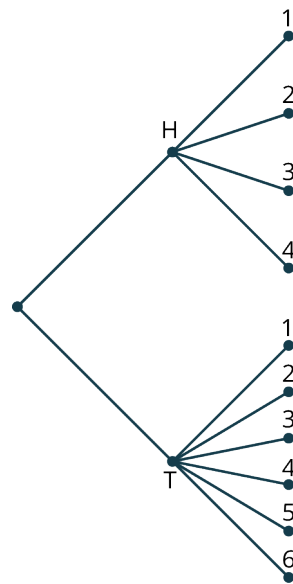


Figure 7.22

Step 4: We can label the leaves to get the sample space ([Figure 7.23](#)).

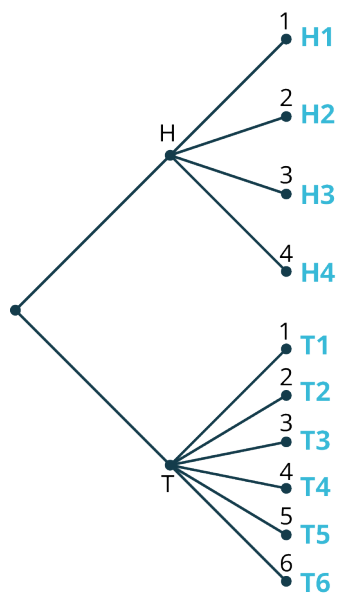


Figure 7.23

The sample space is: {H1, H2, H3, H4, T1, T2, T3, T4, T5, T6}.

- c. **Step 1:** Let's label the trails A, B, and C for ease of labeling. Even though the trails are listed first in the exercise, we can't use the trail choice as our first stage: the trails available to us depend on whether Jess is able to join the trip. So, the first stage is whether Jess joins us (J) or not (N) ([Figure 7.24](#)).



Figure 7.24

Step 2: We list the appropriate trails on each branch ([Figure 7.25](#)).

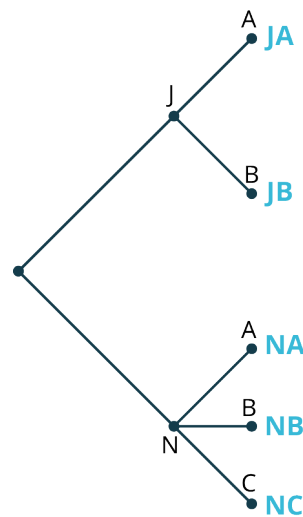


Figure 7.25

So, the sample space is $\{JA, JB, NA, NB, NC\}$.

> YOUR TURN 7.15

1. You have a modified deck of cards containing only $J\heartsuit$, $Q\heartsuit$, and $K\heartsuit$. You draw 2 cards without replacing them (where order matters). Use a tree diagram to identify the sample space.

Check Your Understanding

17. You flip a coin 6 times and note the number of heads. What is the sample space of this experiment?
18. You are ordering a combo meal at a restaurant. The meal comes with either 8 or 12 chicken nuggets, and your choice of crinkle fries, curly fries, or onion rings. Create a table to help you identify the sample space containing your combo meal possibilities.
19. You need one more class to fill out your schedule for next semester. You want to take either History 101 (H), English 220 (E), or Sociology 112 (S). There are two professors teaching the history class: Anderson (A) and Burr (B); one professor teaching the English class: Carter (C); and three people teaching sociology: Johnson (J), Kirk (K), and Lambert (L). Create a tree diagram that helps you identify all your options.
20. Identify the sample space from Exercise 18.
21. Identify the sample space from Exercise 19.



SECTION 7.4 EXERCISES

In the following exercises, you are rolling a special 6-sided die that has both a colored letter and a colored number on each face. The faces are labeled with: a red 1 and a blue A, a red 1 and a green A, an orange 1 and a green B, an orange 2 and a red C, a purple 3 and a brown D, an orange 4 and a blue E. You will roll the die once and take note of something about the face that is showing. Identify the sample space:

1. You note the number.
2. You note the letter.
3. You note the color of the number.
4. You note the color of the letter.
5. You note the number and letter.
6. You note the number and its color.
7. You note the letter and its color.

In Example 7.13, we used a table to identify the possible genetic result when two pea plants are bred; offspring plants get one of the two genes from each parent. Use tables to identify the sample space of offspring for the parents listed in

the following exercises:

8. GG and gg
9. GG and Gg
10. Gg and Gg

You're visiting a pasta bar, where you have your choice of pasta types (cavatappi, ziti, or penne) and sauce (marinara, alfredo, or pesto).

11. Make a table that shows the sample space for your choices.
12. Write out the sample space for the pasta bar.

You have a new art print that you'd like to get framed. You have 3 good choices for the material of the frame: oak, maple, and cherry. You'll also choose a framing mat, for which there are 4 possible colors that will work with your print: plum, lilac, periwinkle, and violet.

13. Make a table that shows the sample space for your choices.
14. Write out the sample space for the framing choices.

You are shopping for a new laptop. The brand you're considering offers laptops with screen sizes 11", 13", 14", 15", and 17". It also offers five memory choices: 4GB, 8GB, 12GB, 16GB, and 32GB.

15. Make a table that shows the sample space for your choices.
16. Write out the sample space for the laptop choices.

For the following exercises, decide whether the described two-stage experiments have independent or dependent stages.

17. Siobhan and Tristan are trying to decide where they will have dinner. Siobhan wants to go to Antoine's, Burger Hut, or the Chowder Palace. Tristan prefers Burger Hut, Chowder Palace, or Duck Duck Taco. They will flip a coin to decide who gets to choose, then that person selects a restaurant from their list.
18. On the TV game show *The Price Is Right*, contestants play games to try to win prizes. One of these games is called "Let 'Em Roll." In this game, players roll five 6-sided dice. These dice each have 3 faces labeled with a car; the other 3 faces are labeled with prize money amounts (one each of \$500, \$1,000, and \$1,500). Players get to roll all 5 dice, and then have the opportunity to win additional rolls. If all 5 dice show a car, the contestant wins a new car. If not, the contestant can use any additional rolls to reroll the dice with prize money showing to try to win the car (or, they can take the total amount of money showing on the dice and end the game). Josh is playing this game, and he has 2 rolls total (these are the two stages of the experiment; the number of cars showing on the dice will be the reported outcome of each stage).
19. In the game Rock, Paper, Scissors, each of 2 players secretly chooses 1 of the 3 title objects. The players reveal their choices simultaneously using hand signals. If the players choose the same object, the game is a tie; if they choose different objects, the winner is determined by the following rules: rock beats scissors, scissors beats paper, paper beats rock. Jim and Eva are gearing up to play the game. The two stages of the experiment are Jim's and Eva's choices.
20. Paul is at the racetrack and is about to place a daily double bet, where he will try to predict the winner of 2 consecutive races. The stages of the experiment are the two races.
21. Mishka is looking for a song to play on a jukebox. The machine requires the user to choose an artist first, then choose a song from among that artist's songs. The stages of the experiment are choosing the artist and choosing the song.

John is playing a game that involves flipping a coin and rolling dice. The coin flip happens first. If the outcome is heads, the player rolls a 4-sided die. If the outcome is tails, the player rolls a 6-sided die.

22. Create a tree diagram to display the possible outcomes of this game.
23. Give the sample space for the game.

In the casino game roulette, a wheel with colored and numbered pockets is spun. At the same time, a marble is spun in the opposite direction in such a way that after a minute or so the marble drops into one of the pockets. Players try to guess the color (red, black, or green) or the number that will appear.

24. Draw a tree diagram that shows the possible color outcomes for 3 spins of the wheel.
25. Give the sample space for the three-stage experiment.

26. Siobhan and Tristan are trying to decide where they will have dinner. Siobhan wants to go to Antoine's, Burger Hut, or the Chowder Palace. Tristan prefers Burger Hut, Chowder Palace, or Duck Duck Taco. They will flip a coin to decide who gets to choose, then that person selects a restaurant from their list. Use an appropriate method to identify the sample space of possible outcomes, which include the person who chooses and the restaurant.

27. On the TV game show *The Price Is Right*, contestants play games to try to win prizes. One of these games is called